

JOINS ASSIGNMENT

SQL Questions Q101–Q150

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-- SECTION 3 – JOINS
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-- Q101. Display customer names and their order IDs using INNER JOIN.
SELECT c.customer_name, o.order_id
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id;

-- Q102. Display customer names, order dates and total amounts.
SELECT c.customer_name, o.order_date, o.total_amount
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id;

-- Q103. Display customer names and order statuses.
SELECT c.customer_name, o.order_status
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id;

-- Q104. Display order IDs and shipping statuses.
SELECT o.order_id, s.shipping_status
FROM orders o
INNER JOIN shipping s
ON o.shipping_id = s.shipping_id;

-- Q105. Display order IDs, shipping cities and delivery dates.
SELECT o.order_id, s.city, s.delivery_date
FROM orders o
INNER JOIN shipping s
ON o.shipping_id = s.shipping_id;

-- Q106. Display order IDs and payment methods.
SELECT o.order_id, p.payment_method
FROM orders o
INNER JOIN payments p
ON o.order_id = p.order_id;

-- Q107. Display order IDs and payment statuses.
SELECT o.order_id, p.payment_status
FROM orders o
INNER JOIN payments p
ON o.order_id = p.order_id;

-- Q108. Display product names and category names.
SELECT p.product_name, c.category_name
FROM products p
INNER JOIN categories c
ON p.category_id = c.category_id;

-- Q109. Display product names and brand names.
SELECT p.product_name, b.brand_name
FROM products p
INNER JOIN brands b
ON p.brand_id = b.brand_id;

-- Q110. Display product names, category names and brand names.
SELECT p.product_name, c.category_name, b.brand_name
FROM products p
INNER JOIN categories c
ON p.category_id = c.category_id
INNER JOIN brands b
ON p.brand_id = b.brand_id;

-- Q111. Display product names and quantities ordered.
SELECT p.product_name, oi.quantity
FROM products p
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INNER JOIN order_items oi
ON p.product_id = oi.product_id;

-- Q112. Display product names and discounts given.
SELECT p.product_name, oi.discount
FROM products p
INNER JOIN order_items oi
ON p.product_id = oi.product_id;

-- Q113. Display order IDs and product names.
SELECT oi.order_id, p.product_name
FROM order_items oi
INNER JOIN products p
ON oi.product_id = p.product_id;

-- Q114. Display customer names, order IDs and payment methods.
SELECT c.customer_name, o.order_id, p.payment_method
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN payments p
ON o.order_id = p.order_id;

-- Q115. Display customer names, product names and quantities ordered.
SELECT c.customer_name, p.product_name, oi.quantity
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN order_items oi
ON o.order_id = oi.order_id
INNER JOIN products p
ON oi.product_id = p.product_id;

-- Q116. Display customer names, product names and category names.
SELECT c.customer_name, p.product_name, cat.category_name
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN order_items oi
ON o.order_id = oi.order_id
INNER JOIN products p
ON oi.product_id = p.product_id
INNER JOIN categories cat
ON p.category_id = cat.category_id;

-- Q117. Display customer names, product names and brand names.
SELECT c.customer_name, p.product_name, b.brand_name
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN order_items oi
ON o.order_id = oi.order_id
INNER JOIN products p
ON oi.product_id = p.product_id
INNER JOIN brands b
ON p.brand_id = b.brand_id;

-- Q118. Display customer names and shipping status.
SELECT c.customer_name, s.shipping_status
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN shipping s
ON o.shipping_id = s.shipping_id;

-- Q119. Display customer names, shipping city and delivery date.
SELECT c.customer_name, s.city, s.delivery_date
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN shipping s
ON o.shipping_id = s.shipping_id;

-- Q120. Display customer names, order amounts and payment status.
SELECT c.customer_name, o.total_amount, p.payment_status
FROM customers c

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INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN payments p
ON o.order_id = p.order_id;

-- Q121. Display all customers and their orders using LEFT JOIN.
SELECT c.customer_name, o.order_id
FROM customers c
LEFT JOIN orders o
ON c.customer_id = o.customer_id;

-- Q122. Display all customers and their order amounts using LEFT JOIN.
SELECT c.customer_name, o.total_amount
FROM customers c
LEFT JOIN orders o
ON c.customer_id = o.customer_id;

-- Q123. Display all products and their order quantities using LEFT JOIN.
SELECT p.product_name, oi.quantity
FROM products p
LEFT JOIN order_items oi
ON p.product_id = oi.product_id;

-- Q124. Display all categories and their products using LEFT JOIN.
SELECT c.category_name, p.product_name
FROM categories c
LEFT JOIN products p
ON c.category_id = p.category_id;

-- Q125. Display all brands and their products using LEFT JOIN.
SELECT b.brand_name, p.product_name
FROM brands b
LEFT JOIN products p
ON b.brand_id = p.brand_id;

-- Q126. Display all orders and their payment details using LEFT JOIN.
SELECT o.order_id, p.payment_method, p.payment_status
FROM orders o
LEFT JOIN payments p
ON o.order_id = p.order_id;

-- Q127. Display all orders and their shipping details using LEFT JOIN.
SELECT o.order_id, s.city, s.shipping_status
FROM orders o
LEFT JOIN shipping s
ON o.shipping_id = s.shipping_id;

-- Q128. Display all products and their category names using RIGHT JOIN.
SELECT p.product_name, c.category_name
FROM products p
RIGHT JOIN categories c
ON p.category_id = c.category_id;

-- Q129. Display all products and their brand names using RIGHT JOIN.
SELECT p.product_name, b.brand_name
FROM products p
RIGHT JOIN brands b
ON p.brand_id = b.brand_id;

-- Q130. Display all orders and their customers using RIGHT JOIN.
SELECT o.order_id, c.customer_name
FROM orders o
RIGHT JOIN customers c
ON o.customer_id = c.customer_id;

-- Q131. Display all payments and their orders using RIGHT JOIN.
SELECT p.payment_id, o.order_id, p.payment_method
FROM payments p
RIGHT JOIN orders o
ON p.order_id = o.order_id;

-- Q132. Display all shipping records and their orders using RIGHT JOIN.
SELECT s.shipping_id, o.order_id, s.shipping_status

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FROM shipping s
RIGHT JOIN orders o
ON s.shipping_id = o.shipping_id;
```

-- Q133. Display pairs of customers who live in the same city using SELF JOIN.

```
SELECT c1.customer_name AS customer_1,
       c2.customer_name AS customer_2,
       c1.city
FROM customers c1
INNER JOIN customers c2
ON c1.city = c2.city
AND c1.customer_id < c2.customer_id;
```

-- Q134. Display pairs of customers from the same city with their email addresses.

```
SELECT c1.customer_name AS customer_1,
       c2.customer_name AS customer_2,
       c1.city
FROM customers c1
INNER JOIN customers c2
ON c1.city = c2.city
AND c1.customer_id < c2.customer_id;
```

-- Q135. Display pairs of customers who live in the same city and have different customer IDs.

```
SELECT c1.customer_id AS customer_1_id,
       c2.customer_id AS customer_2_id,
       c1.city
FROM customers c1
INNER JOIN customers c2
ON c1.city = c2.city
AND c1.customer_id < c2.customer_id;
```

-- Q136. Display customer names and order IDs for customers from Mumbai.

```
SELECT c.customer_name, o.order_id
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
WHERE c.city = 'Mumbai';
```

-- Q137. Display product names and category names for products priced above 10000.

```
SELECT p.product_name, c.category_name
FROM products p
INNER JOIN categories c
ON p.category_id = c.category_id
WHERE p.price > 10000;
```

-- Q138. Display product names and brand names for products with stock greater than 100.

```
SELECT p.product_name, b.brand_name
FROM products p
INNER JOIN brands b
ON p.brand_id = b.brand_id
WHERE p.stock > 100;
```

-- Q139. Display customer names and payment methods for orders above 5000.

```
SELECT c.customer_name, p.payment_method
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN payments p
ON o.order_id = p.order_id
WHERE o.total_amount > 5000;
```

-- Q140. Display customer names and shipping status for delivered shipments.

```
SELECT c.customer_name, s.shipping_status
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN shipping s
ON o.shipping_id = s.shipping_id
WHERE s.shipping_status = 'Delivered';
```

-- Q141. Display customer names and total number of orders using JOIN and GROUP BY.

```
SELECT c.customer_name, COUNT(o.order_id) AS total_orders
FROM customers c
LEFT JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name;
```

```

-- Q142. Display category names and number of products in each category.
SELECT c.category_name, COUNT(p.product_id) AS total_products
FROM categories c
LEFT JOIN products p
ON c.category_id = p.category_id
GROUP BY c.category_id, c.category_name;

-- Q143. Display brand names and average product price.
SELECT b.brand_name, AVG(p.price) AS average_price
FROM brands b
LEFT JOIN products p
ON b.brand_id = p.brand_id
GROUP BY b.brand_id, b.brand_name;

-- Q144. Display customer names and total amount spent by each customer.
SELECT c.customer_name, SUM(o.total_amount) AS total_spent
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name;

-- Q145. Display product names and total quantity sold.
SELECT p.product_name, SUM(oi.quantity) AS total_quantity
FROM products p
INNER JOIN order_items oi
ON p.product_id = oi.product_id
GROUP BY p.product_id, p.product_name;

-- Q146. Display customer names and their payment methods using UNION ALL with shipping cities.
SELECT c.customer_name AS name, c.city AS location
FROM customers c

UNION ALL

SELECT c.customer_name AS name, s.city AS location
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id
INNER JOIN shipping s
ON o.shipping_id = s.shipping_id;

-- Q147. Display customer names and cities using UNION ALL with shipping addresses.
SELECT customer_name AS name, city AS location
FROM customers

UNION ALL

SELECT shipping_address AS name, city AS location
FROM shipping;

-- Q148. Display product names and prices using UNION ALL with order item prices.
SELECT product_name AS item_name, price
FROM products

UNION ALL

SELECT CONCAT('Order Item ', order_item_id) AS item_name, price
FROM order_items;

-- Q149. Display customer names with their city and order status using UNION ALL.
SELECT c.customer_name AS name, c.city AS location
FROM customers c

UNION ALL

SELECT c.customer_name AS name, o.order_status AS location
FROM customers c
INNER JOIN orders o
ON c.customer_id = o.customer_id;

-- Q150. Display product names with their category names using INNER JOIN and include brand names using UNION ALL.
SELECT p.product_name AS item_name, c.category_name AS details
FROM products p
INNER JOIN categories c
ON p.category_id = c.category_id

UNION ALL

```

```
SELECT p.product_name AS item_name, b.brand_name AS details
FROM products p
INNER JOIN brands b
ON p.brand_id = b.brand_id;
```